

Autopsy Forensic Analysis: Procedural Overview

This report details the systematic forensic process conducted using the **Autopsy** digital forensics platform on a **Windows 7 virtual machine**. The analysis followed standard investigative protocols, moving from case initiation to final report generation.

1. Case Initiation and Configuration

The investigative process begins with the creation of a new case file within the Autopsy interface.



Fig. 1: First interface of the software

- **Case Information:** The examiner defines the **Case Name** and the **Base Directory** where all analysis data will be stored.

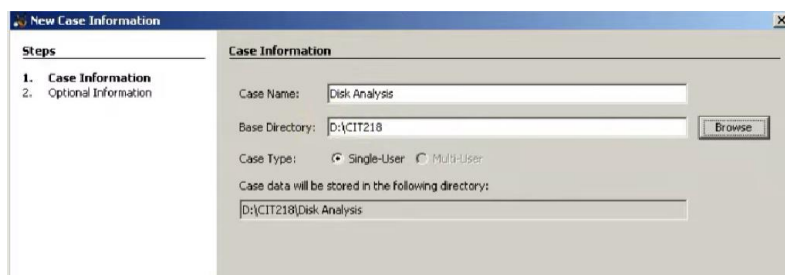


Fig. 2: Adding New Case Information

- **Organization Details:** Optional administrative data, including organization names and point-of-contact information (name, email, and phone), can be documented for chain-of-custody purposes.



Add New Organization

Organization Name:

Point of Contact:

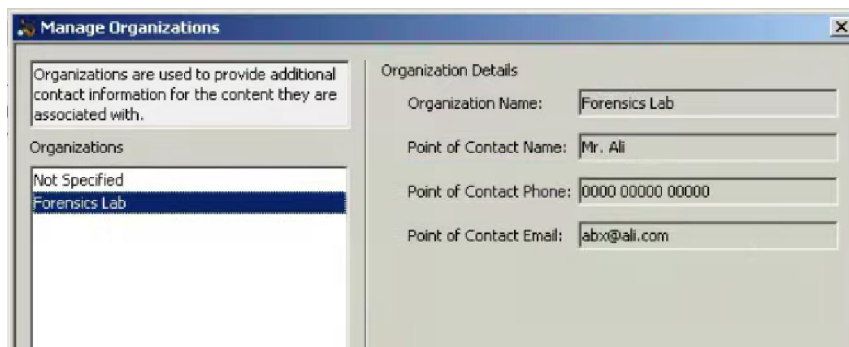
Name:

Email:

Phone:

Fig. 3: Adding Organization Information

- **Examiner Details:** The lead investigator records their name and relevant case numbers before finalizing the initial setup.



Manage Organizations

Organizations are used to provide additional contact information for the content they are associated with.

Organizations

- Not Specified
- Forensics Lab

Organization Details

Organization Name:

Point of Contact Name:

Point of Contact Phone:

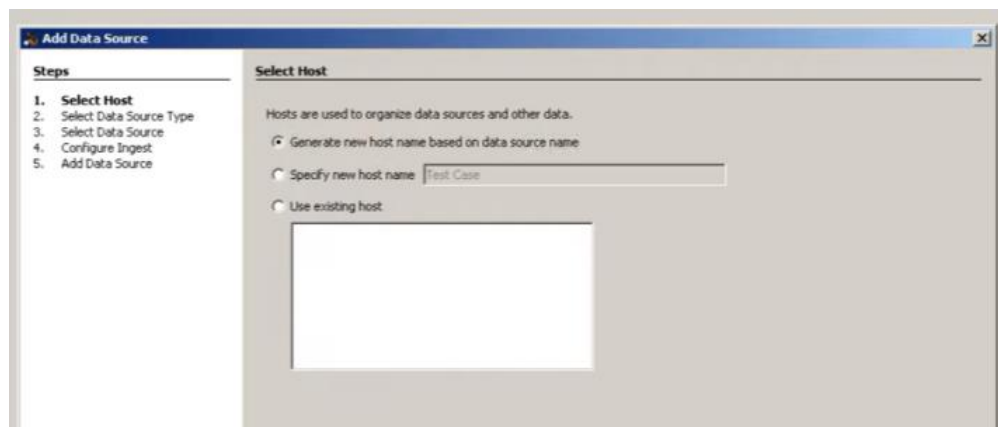
Point of Contact Email:

Fig. 4: Managing Organization Information

2. Data Source Acquisition

Once the case environment is established, the examiner must integrate the evidence for analysis.

- **Host Selection:** A new host name is generated or specified to organize the data sources.



Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Host

Hosts are used to organize data sources and other data.

☒ Generate new host name based on data source name

☐ Specify new host name

☐ Use existing host

Fig. 5: Select Host

- **Source Selection:** In this specific investigation, a **Local Disk** was selected as the primary data source type.

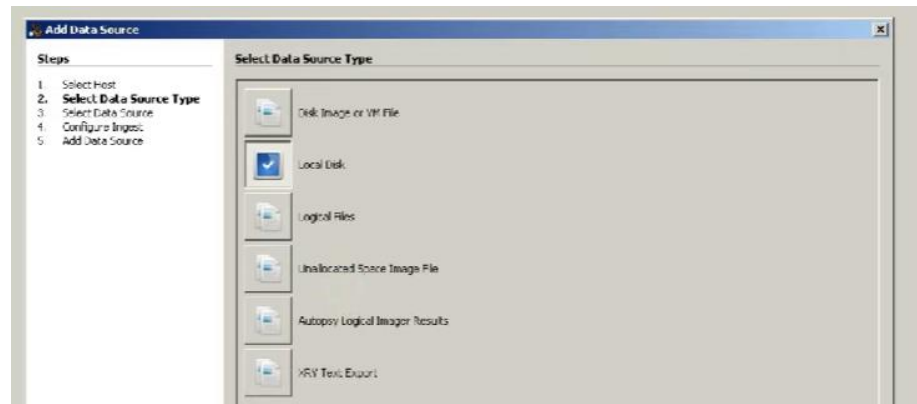


Fig. 6.1: Select data Source Type (in this case local disk)

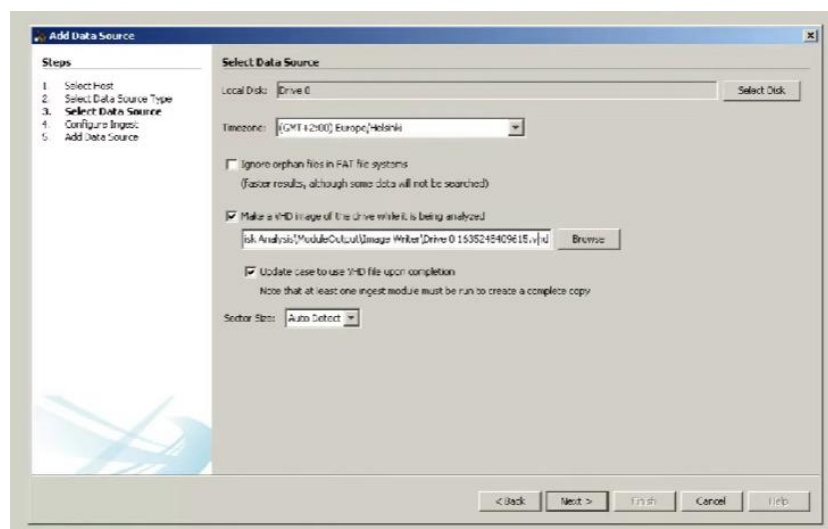


Fig. 6.2: Select Data Source

- **Ingest Modules:** The examiner configures ingest modules to run on all files, directories, and unallocated space. These modules automate the extraction of critical artifacts such as keyword searches, email parsing, and encryption detection.

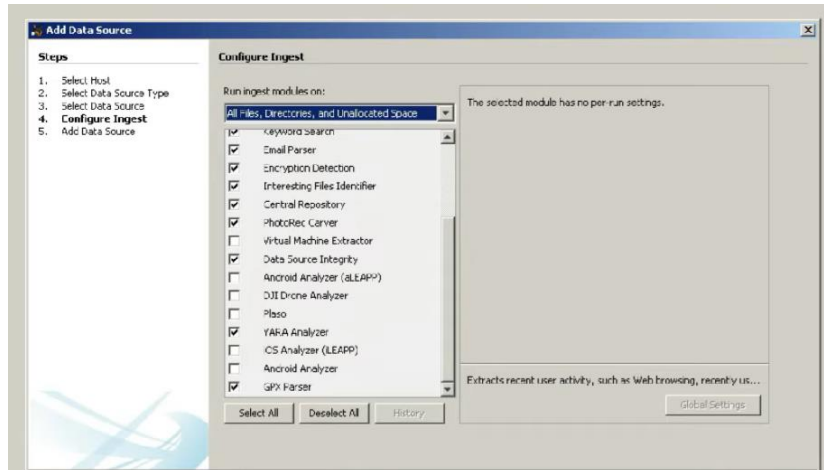


Fig. 7: Configure Ingest

3. Forensic Analysis and Artifact Extraction

Following data ingestion, Autopsy performs a deep analysis of the drive, populating a structured folder system for the examiner to review.

Data Categories and File Views

The software categorizes discovered data into accessible views:

- **File Types:** Files are organized by extension (Images, Videos, Audio, Archives, Databases, etc.).

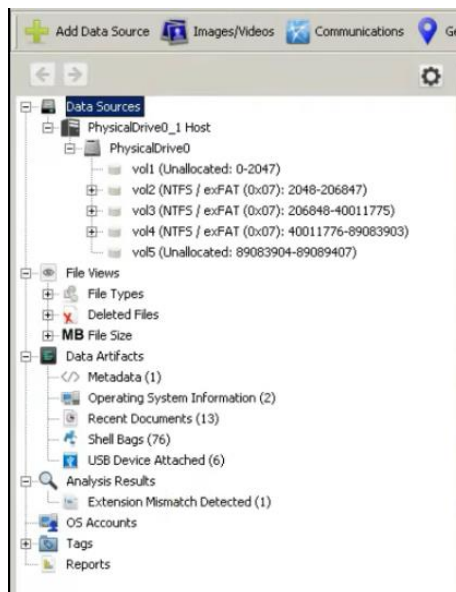


Fig. 8.1 Folder structure

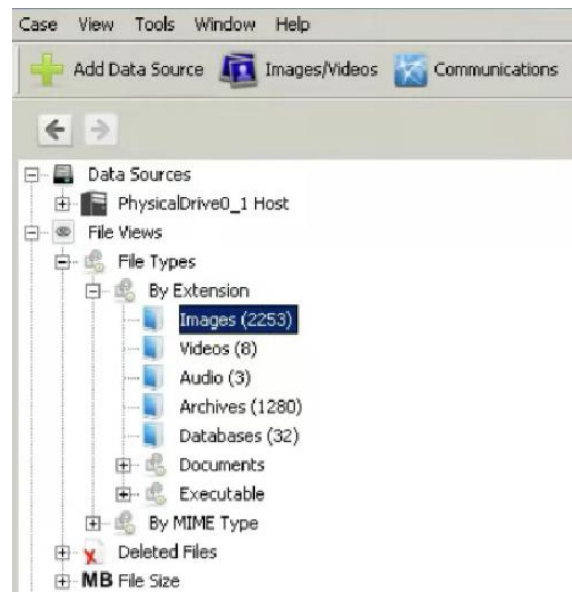


Fig 8.2 Folder structure expanded

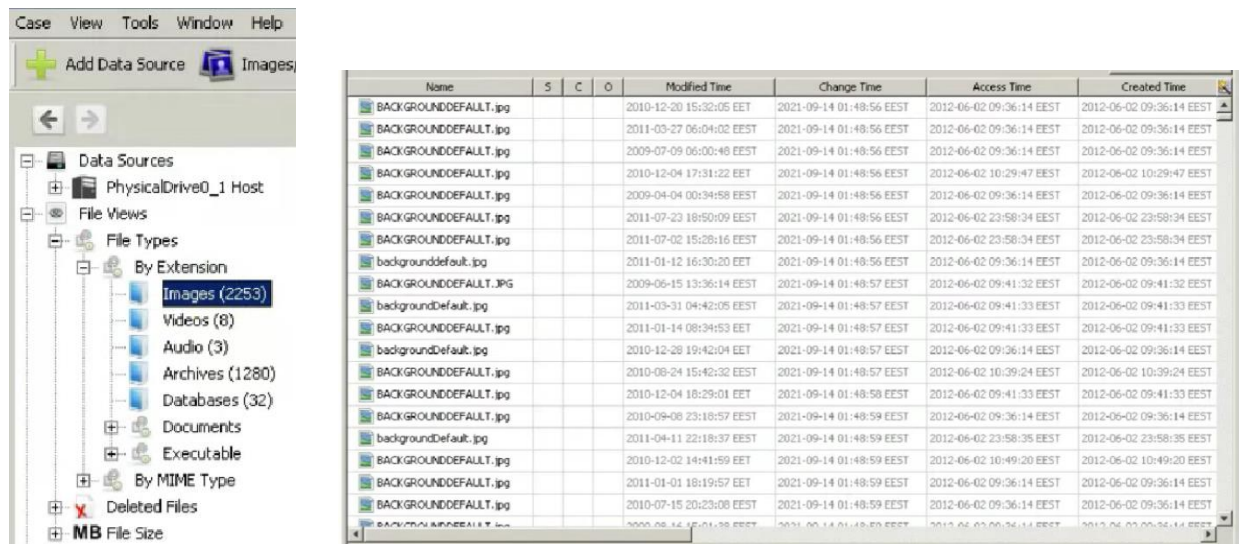


Fig. 9.1, 9.2: Images Details

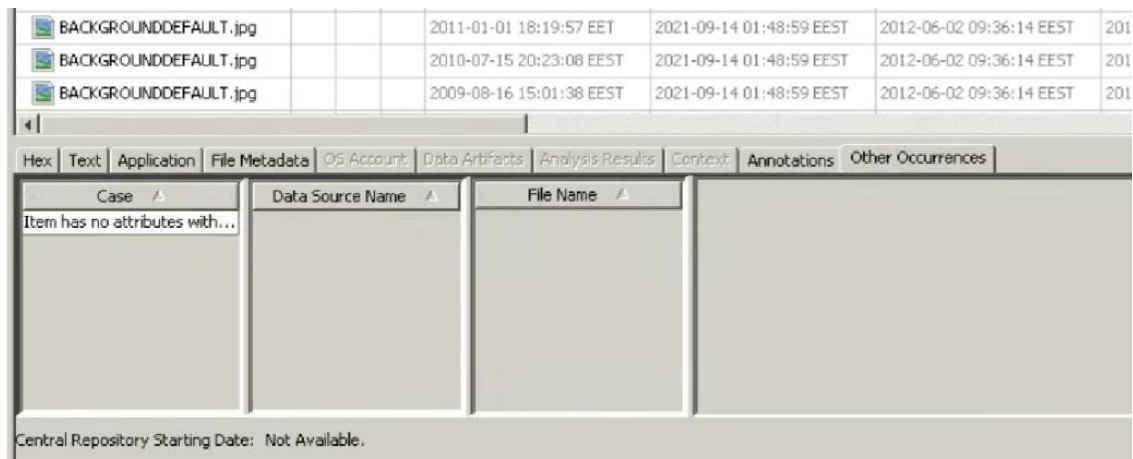


Fig. 9.3 Particular Image's Detail

- **Error Reporting and Messages:** A dedicated message icon provides access to status updates. These messages are essential for troubleshooting issues related to information extraction.

Module	Num	New?	Subject	Timestamp
Hash Lookup	1	●	No notable hash set.	2021/10/26 14:00:00
Hash Lookup	1	●	No known hash set.	2021/10/26 14:00:00
Recent Activity	1	●	Started PhysicalDrive0	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.python.proxies.GPX_Parser_Mo...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.python.proxies.module\$Android...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.centralreposit...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.keywordsearc...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.data...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.dron...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.embe...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.encr...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.filety...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.hash...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.inter...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.leapp...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.leapp...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.pictu...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.vmx...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.yara...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.recentactivity...	2021/10/26 14:00:00
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.thunderbirdna...	2021/10/26 14:00:00

Fig. 10: Messages

- **Deleted Files:** The system identifies and segregates recently deleted files from the file system for further scrutiny.

File Name	Size	Modified	Changed	Access	Created
Local State~Rfa2922.TMP	0	2021-10-25 11:04:01 EEST	2021-10-25 11:04:01 EEST	2021-10-25 11:04:01 EEST	2021-10-25 11:04:01 EEST
Local State~Rfa248s.TMP	0	2021-10-25 11:03:46 EEST	2021-10-25 11:04:00 EEST	2021-10-25 11:03:46 EEST	2021-10-25 11:03:46 EEST
62f4d3ed-4fe4-476-9275-f52d66a1d93.tmp	0	2021-10-25 11:03:35 EEST	2021-10-25 11:03:35 EEST	2021-10-25 11:03:35 EEST	2021-10-25 11:03:35 EEST
f76d3ab-4ab3-47c8-aa10-bcd30c4798d7.tmp	0	2021-10-25 11:03:42 EEST	2021-10-25 11:03:42 EEST	2021-10-25 11:03:42 EEST	2021-10-25 11:03:42 EEST
de5e729e-7fb1-49e1-b7f9-c51c53917042.tmp	0	2021-10-25 11:03:45 EEST	2021-10-25 11:03:45 EEST	2021-10-25 11:03:45 EEST	2021-10-25 11:03:45 EEST
820c4e99-1345-4871-aa92-4691d9edde03.tmp	0	2021-10-25 11:03:48 EEST	2021-10-25 11:03:48 EEST	2021-10-25 11:03:48 EEST	2021-10-25 11:03:48 EEST
a4d5c1e-a5ee-405e-0690-a3491cca7399.tmp	0	2021-10-25 11:03:49 EEST	2021-10-25 11:03:49 EEST	2021-10-25 11:03:49 EEST	2021-10-25 11:03:49 EEST
5d96d521-ds238c3.customDestinations-ms-Rfa14bf.TI	0	2021-10-25 11:03:36 EEST	2021-10-25 11:03:55 EEST	2021-10-25 11:03:36 EEST	2021-10-25 11:03:36 EEST
5d96d521-ds238c3.customDestinations-ms-Rfa1c9f.TI	0	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST
5d96d521-ds238c3.customDestinations-ms-Rfa14bf.TI	0	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST	2021-10-25 11:03:55 EEST

Fig. 11: Recently deleted files

- **Metadata:** Detailed timestamps (Modified, Changed, Access, and Created times) are provided for every identified file.

User Activity and Web Artifacts

The examiner can reconstruct user behavior by analyzing several data artifacts:

- **Recent Activity:** Includes recently accessed documents and programs currently or previously run on the system.

Source Module Name	Report Name	Created Time	Report File Path
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:11:23 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:11:23 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Users/Administrat...	2021-10-25 11:11:29 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Users/Default/NT...	2021-10-25 11:11:31 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/ServiceP...	2021-10-25 11:11:32 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/ServiceP...	2021-10-25 11:11:33 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:11:34 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Users/Administrat...	2021-10-25 11:11:35 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:11:41 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:12:02 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:12:03 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:12:03 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
RecentActivity	RegRipper /mg_PhysicalDrive0/vol_vol3/Windows/System3...	2021-10-25 11:12:04 EEST	D:\CIT218\Disk Analysis\ModuleOutput\RecentActivity\reg_1...
HTML Report		2021-10-25 14:59:29 EEST	D:\CIT218\Disk Analysis\Reports\Disk Analysis HTML Report :...

Fig. 12: Recent Activity

- **Web History:** Comprehensive logs of web cache, cookies, downloads, and search history are extracted.

Source File	S	C	O	URL	Date Accessed	Referrer URL
History	2			https://www.bbc.co.uk/	2021-10-25 11:03:19 EEST	https://www.bbc.co.uk/
History	2			https://www.arabnews.com/	2021-10-25 11:03:19 EEST	https://www.arabnews.com/
History	2			https://www.bbc.com/	2021-10-25 11:03:28 EEST	https://www.bbc.com/
index.dat	3			https://www.google.com/images/branding/googlelogo/1s/...	2021-10-25 07:42:16 EEST	
index.dat	3			https://www.google.com/images/branding/googlelogo/1s/...	2021-10-25 07:42:13 EEST	
index.dat	3			https://www.google.com/chrome/itabc/images/thank-you/...	2021-10-25 07:42:39 EEST	
index.dat	3			https://www.google.com/images/branding/searchlogo/1s/...	2021-10-25 07:42:20 EEST	
index.dat	3			https://www.google.com/?gws_rd=ssl	2021-10-25 07:42:13 EEST	
index.dat	3			https://www.google.com/	2021-10-25 07:42:17 EEST	
index.dat	3			https://www.google.com/gtm.js?id=GTM-226T...	2021-10-25 07:42:36 EEST	
index.dat				iecompab:bolaw.com	2021-10-25 07:45:13 EEST	
index.dat				iecompab:clcrs.com.br	2021-10-25 07:45:13 EEST	
index.dat				iecompab:dailytelegaph.com	2021-10-25 07:45:13 EEST	
index.dat				iecompab:duba.net	2021-10-25 07:45:13 EEST	
index.dat				iecompab:iterahernandezsarcia.com	2021-10-25 07:45:13 EEST	

Fig. 13: Web History

- **Shell Bags:** These are analyzed to identify which folders were accessed by the user, even if those folders no longer exist.

Source File	S	C	O	Path	Key	Last Modified
UserClass.dat				My Computer	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-10-25
UserClass.dat				My Computer(D:\)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-10-25
UserClass.dat				My Computer(D:\CT2\8)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-10-25
UserClass.dat				My Computer(D:\CT2\8)\Disk Analysis	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-10-25
UserClass.dat				My Computer(D:\CT2\8)\logical_Inlager_Forensics_20211...	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-10-25
UserClass.dat				My Computer(I:\)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(I:\OSForensics Pro 6.0 Build 1300)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(I:\OSForensics Pro 6.0 Build 1300)\crack	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(I:\AccessData Registry Viewer 2.0)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(C:\)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(C:\ProgramData)	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13
UserClass.dat				My Computer(C:\ProgramData)\Microsoft	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\...	2021-09-13

Fig. 14: Shell Bags

Advanced Investigation Results

The analysis identifies potential areas of interest through automated flags:

- **Encryption Suspected:** Files that appear encrypted are segregated for deeper investigation.



Fig. 15.1, 15.2: Encrypted files

- **Keyword Hits:** The system lists email addresses and specific literal or regular expression matches found across various software.
- **User Content Suspected:** Suspicious files are categorized by content type for efficient manual review.

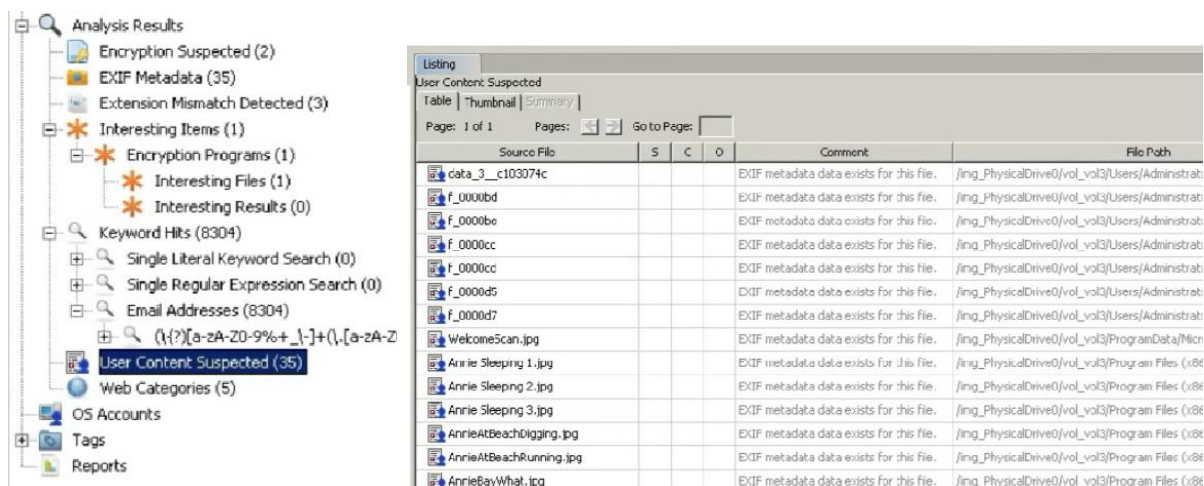


Fig. 16.1, 16.2: User Content Suspected

- **OS Accounts:** The software extracts details regarding login names (e.g., Administrator, Guest), login counts, and home directories.

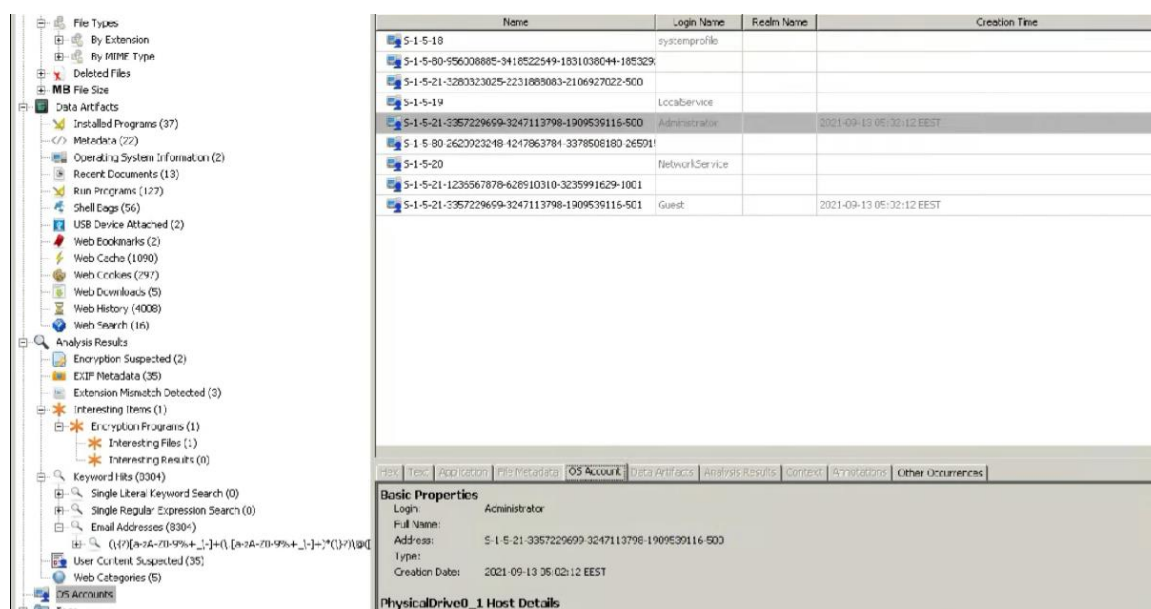


Fig. 17.1: OS Accounts

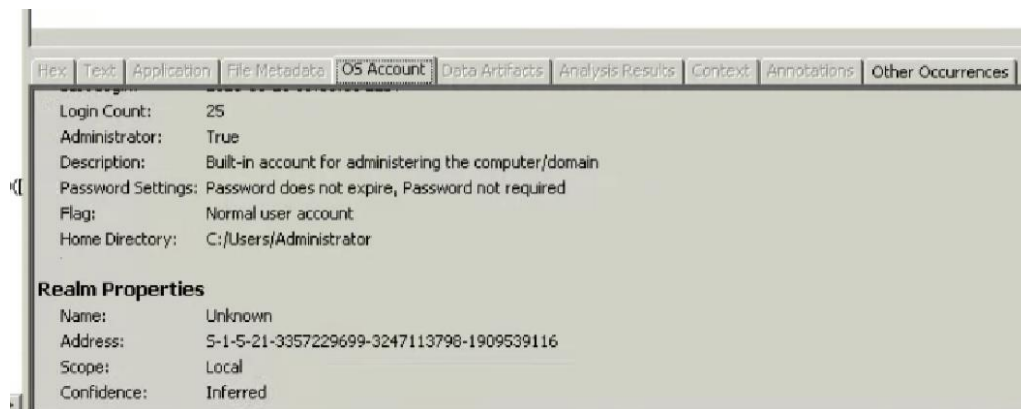


Fig. 17.2: Particular OS Accounts Details

4. Timeline and Data Summary

To establish a chronological sequence of events, the **Timeline** tool visualizes file activity and events over specific years, months, or days. A comprehensive **Data Source Summary** provides a high-level overview of the drive, including a breakdown of file types (Documents, Executables, Images, etc.) and the overall ingest status.

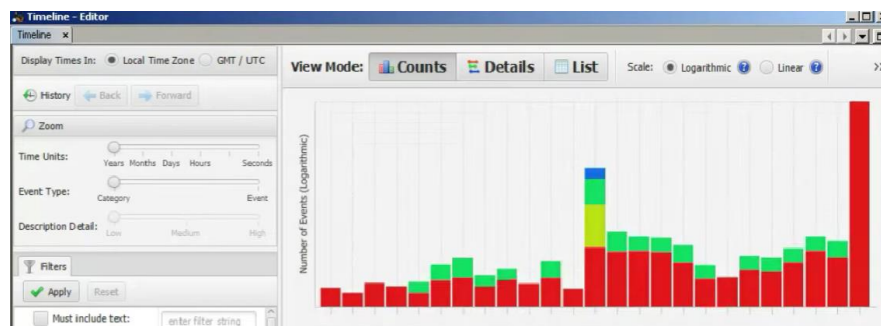


Fig. 18: Timeline

Data Source Name	Ingest Status	Type	Files	Results	Tags
PhysicalDrive0	Started	OS Drive (Windows 7 Ultimate S...	108177	15262	0

Types: User Activity | Analysis | Recent Files | Past Cases | Geolocation | Timeline | Ingest History | Container | Export

Usage: OS Drive (Windows 7 Ultimate Service Pack 1)

Fig. 19.1: Data Source Summary

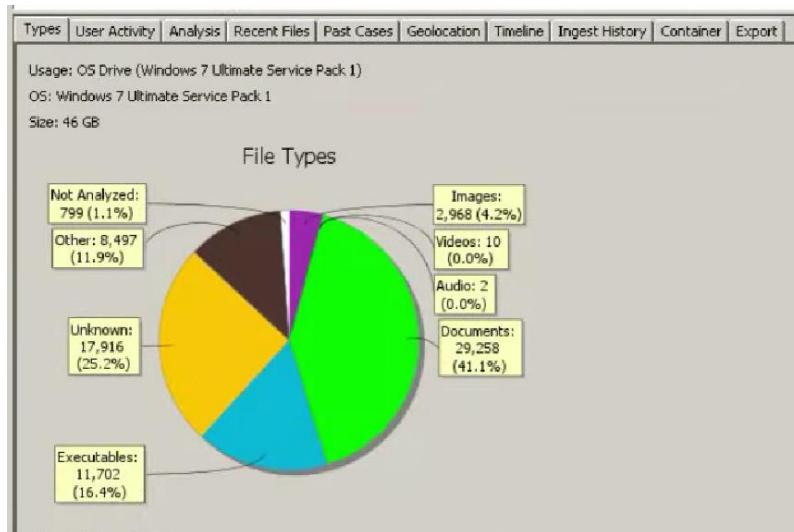


Fig. 19.2: Data Source Summary

5. Final Reporting

The final stage of the investigation involves consolidating all findings into a formal document.

- **Format Selection:** The examiner can generate reports in several formats, including **Excel**, **Text**, or **HTML**.

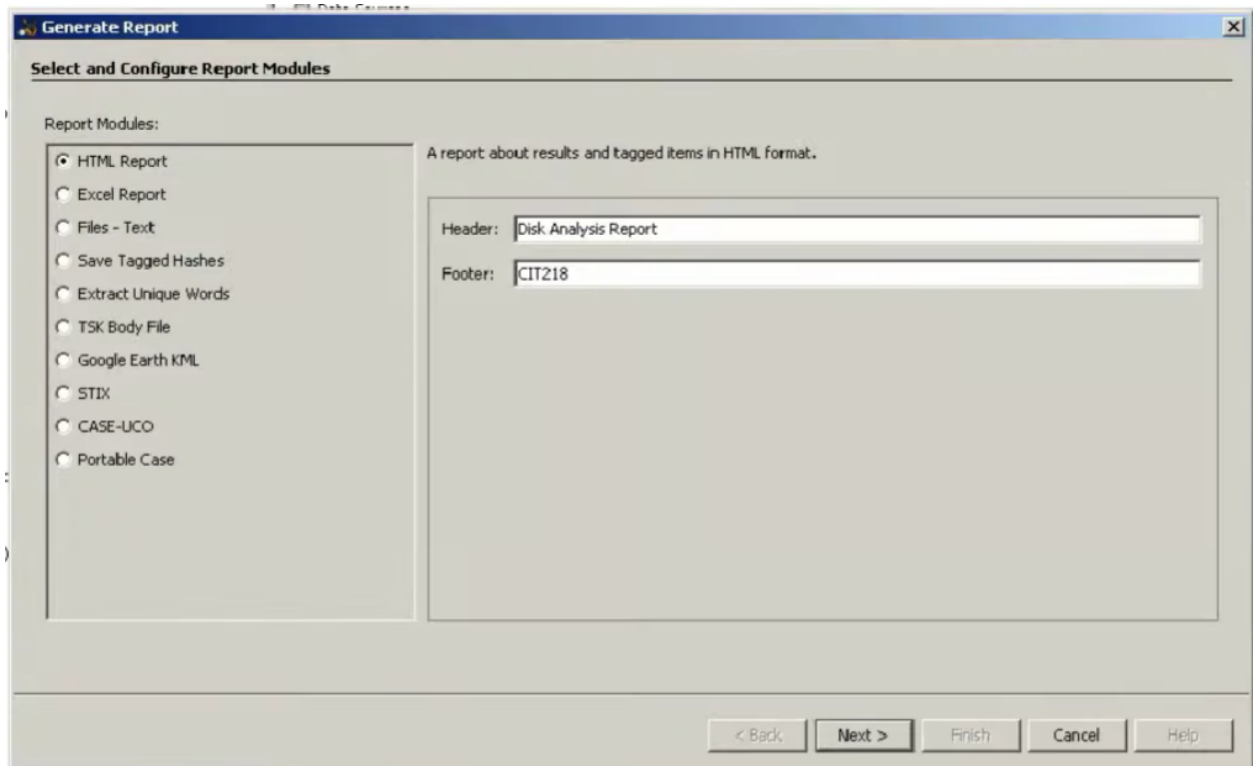


Fig. 20: Select and Configure Report Modules

- **Configuration:** For this case, an **HTML report** was selected. The examiner can customize headers and footers to meet jurisdictional or organizational requirements.

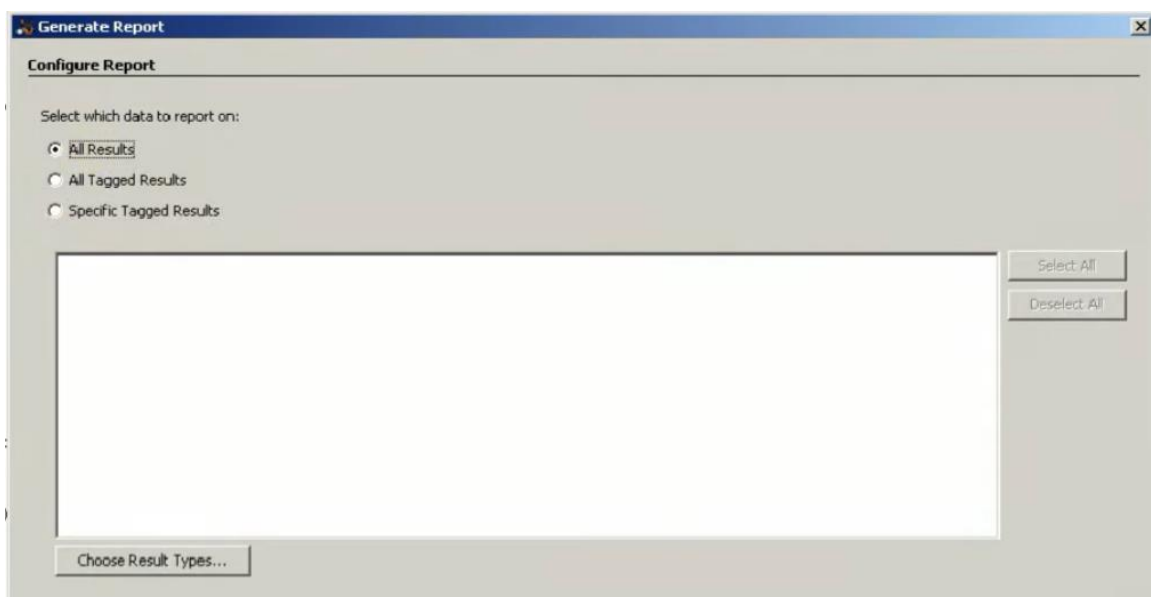


Fig. 21: Select and Configure Report Modules

- **Export:** The resulting report includes all relevant details from the investigation—including case summaries, EXIF metadata, and flagged items—ready for viewing in a standard web browser.

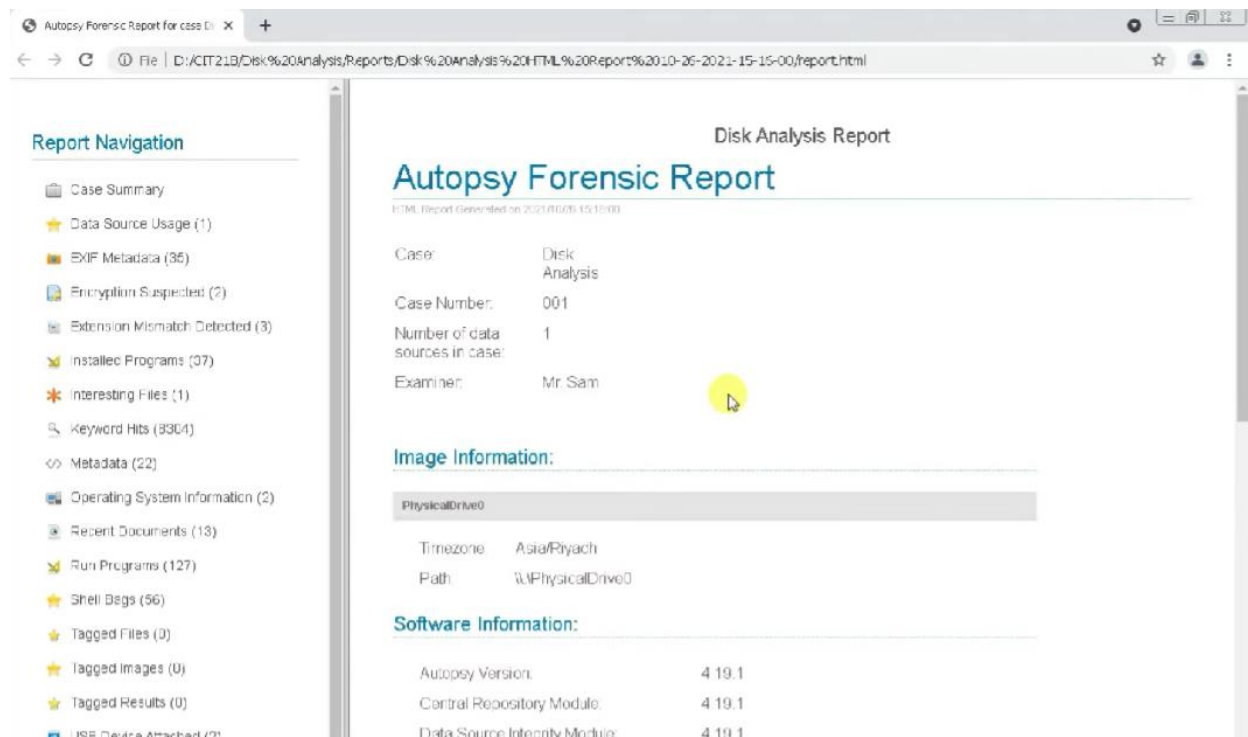


Fig. 22: Generated HTML-format Report